DATASHEET

EETX433A

Multifunctional RF Transmitter

Low-power RF solution for wireless remote control

TRANSMIT POWER 13 dBm	MODULATION OOK / ASK
SLEEP CURRENT 800 nA	TRANSMIT CURRENT 12 mA

VERSION	PACKAGE INFORMATION	DEVELOPED BY	REVISION
v1.0	SOP8	Engeletron Engenharia	v1.1 · 2026-06-02

Features

Transmit power	13 dBm
Modulation	OOK / ASK
Sleep current	800 nA
Transmit current	12 mA
Frequencies	433.92 MHz
Supply voltage	2.0 V – 3.6 V
Package	SOP-8
Certifications	FCC, CE, ANATEL

Applications

- Security access control systems
- Wireless alarms
- Fire detection systems
- Wireless remote controls
- Home automation
- Ultra-low-power IoT sensors

Overview

The EETX433A integrates a 433 MHz PLL oscillator, an RF transmitter, and a microcontroller (8-bit MCU core) on a single chip, delivering a simpler design, more flexible encoding, lower cost, and a smaller footprint — fully compliant with FCC, CE and ANATEL standards.

A built-in key-detection auto-shutdown feature minimizes standby current. The device also integrates output overcurrent protection, on-chip over-temperature protection, and supply undervoltage protection.

Building a complete wireless transmission system with the EETX433A requires only 2 inductors, 3 capacitors, resistors, a battery, and buttons — a fully integrated solution for remote control applications.

Functional Block Diagram

The EETX433A integrates on a single die all blocks required for a standalone 433.92 MHz RF transmitter: reference oscillator, PLL for carrier synthesis, OOK/ASK modulator, power amplifier, 8-bit MCU with code ROM, power-management (UVLO/OTP/OCP), and operating-mode selection logic (CFG).

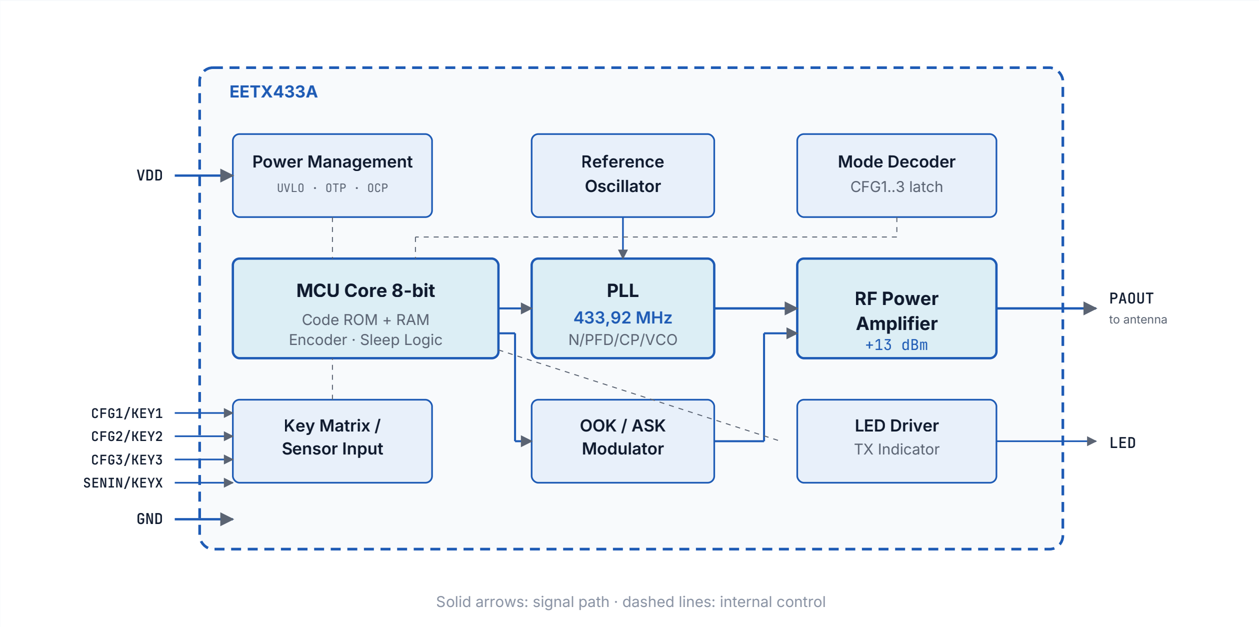


Figure 0: EETX433A internal functional block diagram

Functional Operation

This section describes the operational behavior of the EETX433A for designers using the part. All parameters listed below are fixed in the embedded MCU firmware and cannot be altered in production.

Factory-Programmed Address

The 20-bit address is programmed at the factory during EETX433A production and is unique per order code. No additional programming step is required on the customer PCB — the IC is delivered ready-to-use, in a 4 000-piece reel sharing the same address. To order batches with different addresses, contact Engeletron.

Transmission Trigger

Mode 0 (Transmitter): transmission triggers when any KEY pin (CFG1..3 or SENIN) is connected to VDD (logic 1) — typically via a momentary key between the pin and VDD. Transmission continues while the key is held, with a minimum duration of 800 ms (16 packets) and a maximum of 5000 ms (100 packets). If the user releases the key before 800 ms, the IC still completes the minimum duration.

Modes 1–7 (Sensor): transmission triggers on the rising edge (0 → 1 transition) of the SENIN pin, coming from a digital sensor with ON/OFF output (magnetic TMR or digital PIR sensor). Transmission duration is fixed at 1 200 ms (24 packets) regardless of the sensor state after the trigger.

PIR Mode Re-arming (Motion)

In PIR (Passive) modes, after a transmission the IC enters a lockout state until the sensor remains at logic low (no motion detected) for at least 120 seconds (2 minutes) continuously. This interval prevents transmission bursts in environments with continuous motion (e.g. wind-blown curtains). Only after this idle period will the IC accept a new SENIN rising edge and trigger a new transmission.

CFG Pin Sampling

CFG1, CFG2 and CFG3 pins are sampled ONLY at power-up (VDD ramp from 0 V to the operating level). Dynamic changes during operation are ignored. After the latch (≤ 1 ms), the same pins become KEY1/KEY2/KEY3 (mode 0) or are reserved for internal use (modes 1–7).

Key Mapping (Mode 0)

In Mode 0 (Transmitter), the EETX433A always operates as a matrix of up to 6 keys. CFG1, CFG2 and CFG3 pins must be left floating (no connection to VDD or GND) at power-up — this is what selects Mode 0. The designer chooses how many physical keys to use (from 1 to 6) per the application schematics. Each key, when pressed, transmits a unique value in the 4-bit data field:

Key	D3	D2	D1	D0	Decimal value
KEY1	0	0	0	1	1
KEY2	0	0	1	0	2
KEY3	0	0	1	1	3
KEY4	0	1	0	0	4
KEY5	0	1	0	1	5
KEY6	0	1	1	0	6

SENIN Pin (Sensor Input)

In modes 1–7, the SENIN pin operates as a pure digital input (high impedance). There is no internal pull-up or pull-down: the connected sensor is responsible for maintaining the defined logic level (high or low) under all operating conditions. Sensors with push-pull outputs (e.g. digital TMR, ON/OFF digital PIR) meet this requirement without additional components. For open-drain output sensors or dry contacts, use an external pull-down resistor (typical 100 kΩ → GND) to guarantee a defined low level at rest.

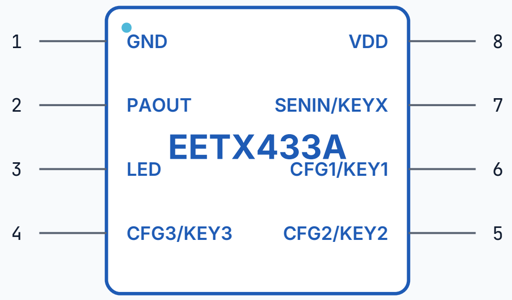
Operational Timing Parameters

Parameter	Symbol	Min.	Typ.	Max.	Unit
Tempo de wake-up (power-up até 1º bit)	t_WK	–	50	100	ms
Tempo de estabilização do PLL	t_PLL	–	0,2	1	ms
Duração de 1 pacote	t_PKT	–	50	–	ms
Modo 0 — TX mínimo (16 pacotes)	t_TX0_min	–	800	–	ms
Modo 0 — TX máximo (100 pacotes)	t_TX0_max	–	–	5000	ms
Modos 1–7 — TX fixo (24 pacotes)	t_TX1_7	–	1200	–	ms
Rearme PIR — sem movimento contínuo	t_REARM	120	–	–	s
Latch dos pinos CFG (power-up)	t_CFG	–	–	1	ms

Compatible Receiver

To receive and decode the signal transmitted by the EETX433A, an ASK (Amplitude Shift Keying) RF receiver with a reception bandwidth (Δf_RX) of ±300 kHz or higher is recommended — comfortably covering the ±250 kHz unit-to-unit f_0 variation of the EETX433A. Recommended sensitivity is -110 dBm to -112 dBm for typical residential/commercial range. Frame decoding (24λ pilot + 20-bit address + 4-bit data + 4λ end, λ = 500 µs) is typically performed by a microcontroller on the receiver side.

Pinout & Pin Configuration



SOP-8 (top view)

Pin Description

Pin	Name	I/O	Function
1	GND	I/O	Power ground reference.
2	PAOUT	O	Power amplifier (PA) output.
3	LED	O	Data transmission indicator (LED driver).
4	CFG3/KEY3	I	Input-only pin. Operation mode and sensor type configuration 3; Direct Key 3 and/or Key 6 associated with the transmitter key matrix.
5	CFG2/KEY2	I	Input-only pin. Operation mode and sensor type configuration 2; Direct Key 2 and/or Key 5 associated with the transmitter key matrix.
6	CFG1/KEY1	I	Input-only pin. Operation mode and sensor type configuration 1; Direct Key 4 and/or Key 6 associated with the transmitter key matrix.
7	SENIN/KEYX	I/O	Input or output pin. Digital sensor input; Transmitter key matrix.
8	VDD	I/O	Positive supply voltage.

Configuration Modes (CFG Pins)

Beyond operating as a transmitter with up to 6 keys, the EETX433A can be configured as a magnetic opening sensor (door/window) or a passive PIR motion sensor. Configuration is set by tying pins CFG1, CFG2 and CFG3 to GND (level 0) or VDD (level 1) per the table below:

Pins CFG1, CFG2 and CFG3 are sampled in hardware ONLY at power-up. The binary combination of the three pins (weight CFG3:CFG2:CFG1) selects one of 8 operating modes. Dynamic changes during operation are ignored. The SENIN pin becomes the external sensor input in modes 1–7. See "Functional Operation" for details.

- **Transmitter Mode (Mode 0)** — All three CFG pins are left open during power-up (no connection to VDD/GND) — this selects Mode 0. The IC always operates as a 6-key matrix; the designer chooses how many physical keys to fit (from 1 to 6) per the schematics. Each key transmits a unique 4-bit value (see "Key Mapping").
- **Magnetic Mode (Modes 1, 2, 3, 6)** — The EETX433A monitors the SENIN pin connected to a magnetic sensor (e.g. TMR, Reed switch). Ideal for door/window opening detection.
- **Passive / PIR Mode (Modes 4, 5, 7)** — The EETX433A monitors the SENIN pin connected to a pyroelectric presence sensor (PIR). Must be a DIGITAL pyrosensor with ON/OFF (high/low) output. Ideal for motion detection.

MODE	TYPE	TRANSMITS	CFG3	CFG2	CFG1	SENIN
0	TRANSMISSOR	TECLA	Aberto	Aberto	Aberto	Aberto
1	MAGNÉTICO	V1B3	0	0	1	Entrada
2	MAGNÉTICO	V3B1	0	1	0	Entrada
3	MAGNÉTICO	A4F5B3	0	1	1	Entrada
4	PASSIVO	V1B3	1	0	0	Entrada
5	PASSIVO	V3B1	1	0	1	Entrada
6	MAGNÉTICO	V2B1	1	1	0	Entrada
7	PASSIVO	V2B1	1	1	1	Entrada

Legend: V = Violation · B = Low battery · A = Open · F = Close. Notation: the number after each letter is the 4-bit value (D3D2D1D0) transmitted in the data field when that event occurs. Example — Mode 1 (V1B3): on a Violation the transmitter sends data = 0001 (= 1); when the sensor reports Low Battery, data = 0011 (= 3) is sent. The receiver decodes the 4-bit field to identify the event.

Block Diagram & Typical Application

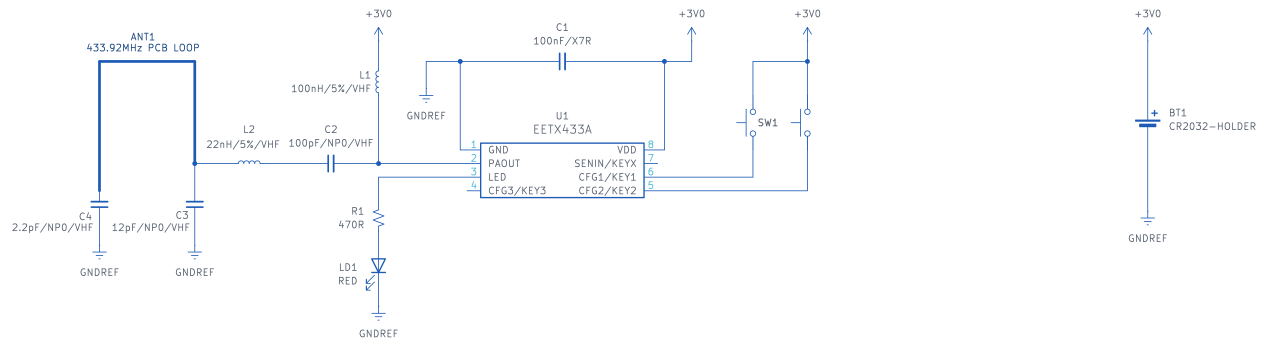


Figure 1: 433.92 MHz typical application — 2 buttons

Block Diagram & Typical Application

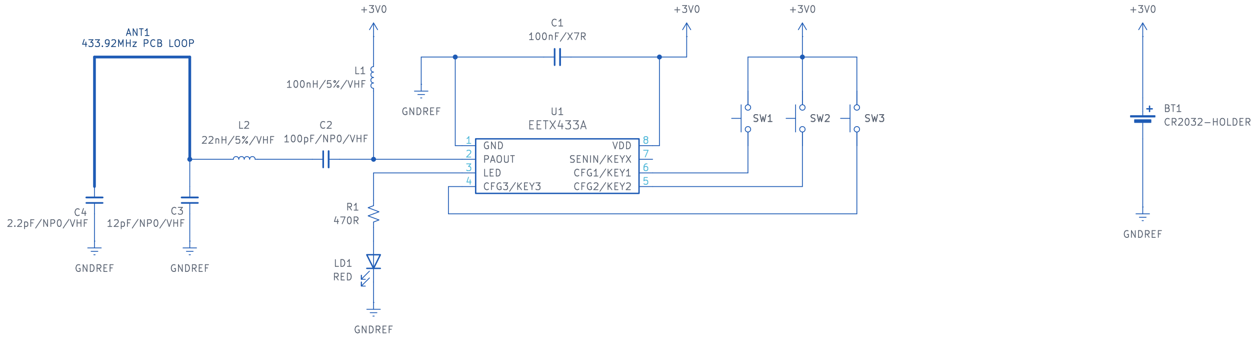


Figure 2: 433.92 MHz typical application — 3 buttons

Block Diagram & Typical Application

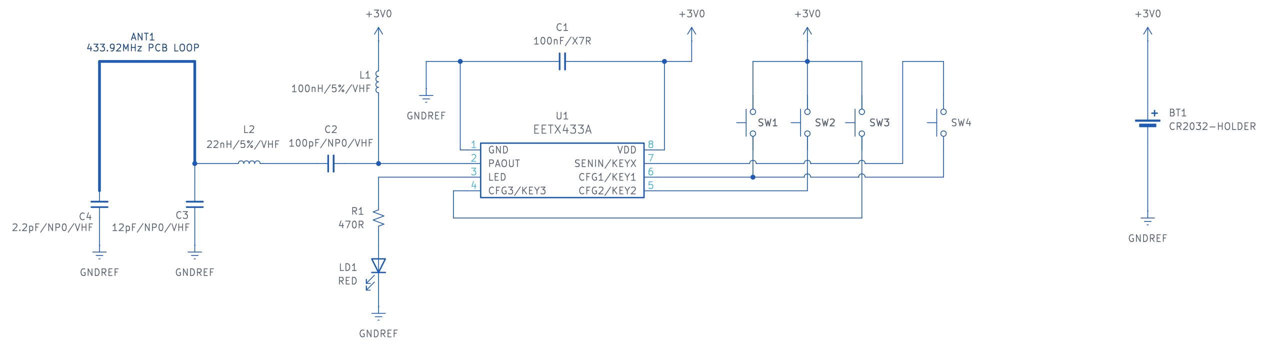


Figure 3: 433.92 MHz typical application — 4 buttons

Block Diagram & Typical Application

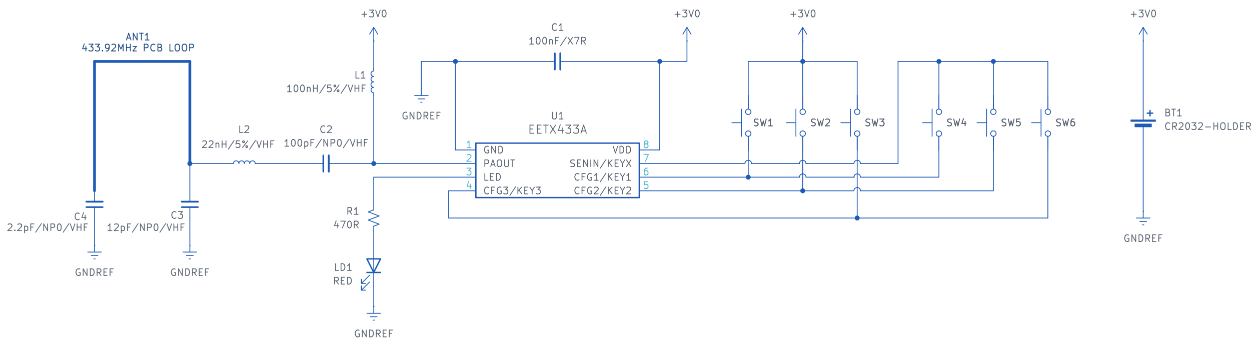


Figure 4: 433.92 MHz typical application — 6 buttons

Block Diagram & Typical Application

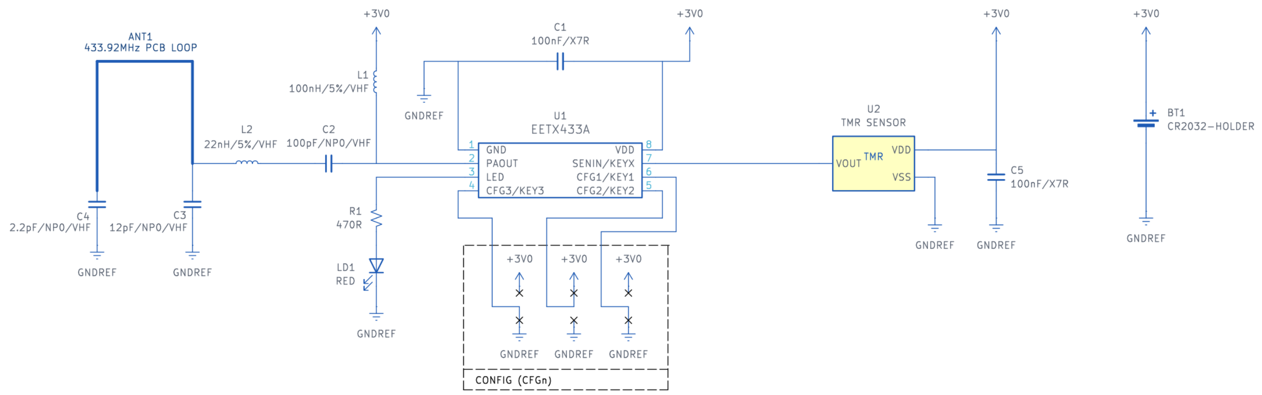


Figure 5: Typical application as magnetic opening sensor (TMR + EETX433A)

Block Diagram & Typical Application

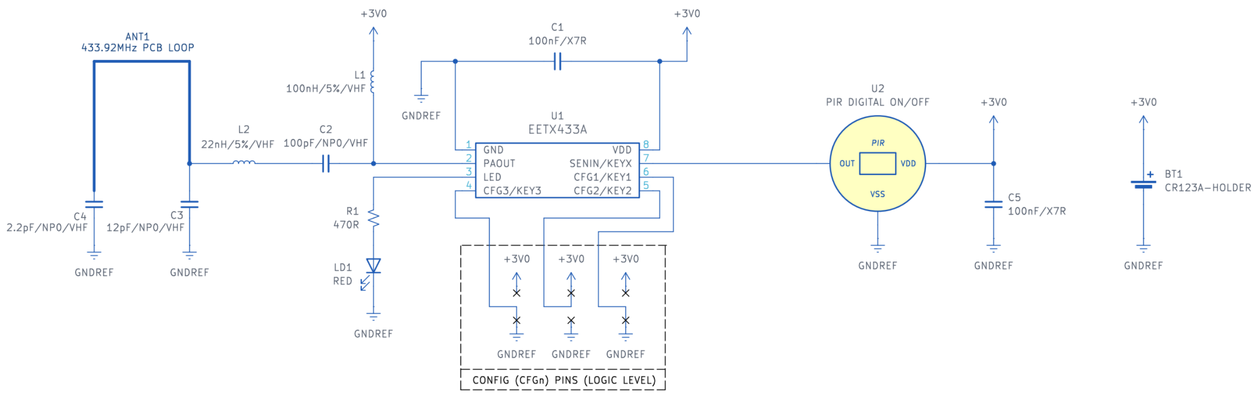


Figure 6: Typical application as motion sensor (digital ON/OFF PIR + EETX433A)

Block Diagram & Typical Application

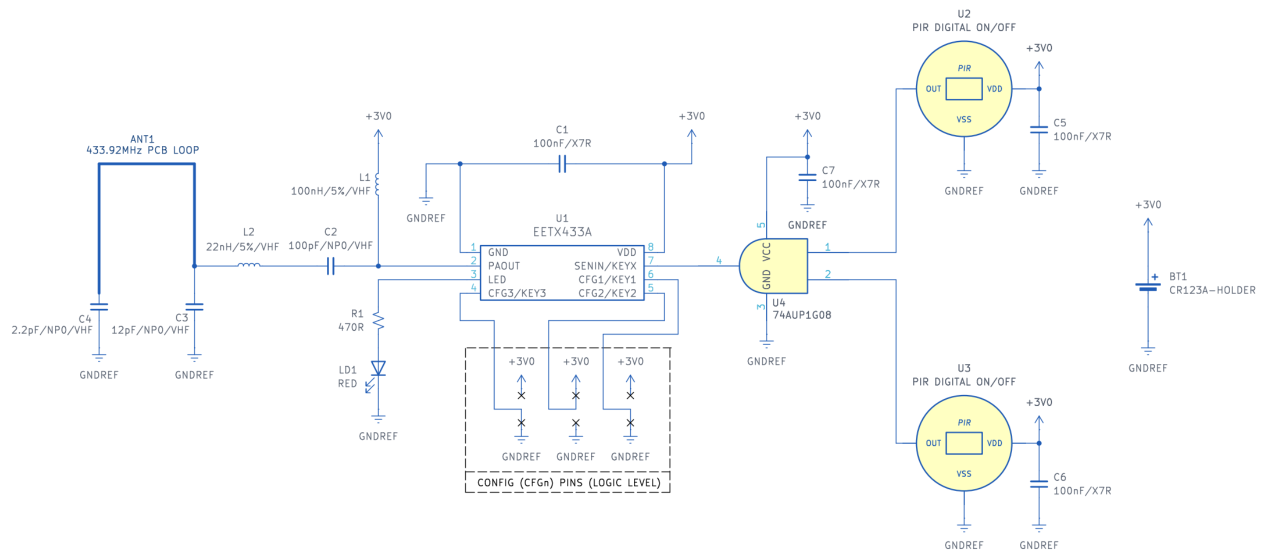


Figure 7: Typical application as dual-lens motion sensor (2× digital PIR + ultra-low-power AND logic gate + EETX433A)

Block Diagram & Typical Application

Remote actuation via vehicle-light pulse: the module is connected in parallel with the high-beam headlight. The user must keep the high beam asserted for at least 2 seconds to guarantee complete transmission — this covers wake-up (≈ 50 ms) + PLL settling (≈ 0.2 ms) + fixed 1 200 ms transmission + margin. The bridge rectifier + 3.3 V LDO make the circuit insensitive to battery polarity (12 V or 24 V) and avoid the need for an internal battery.

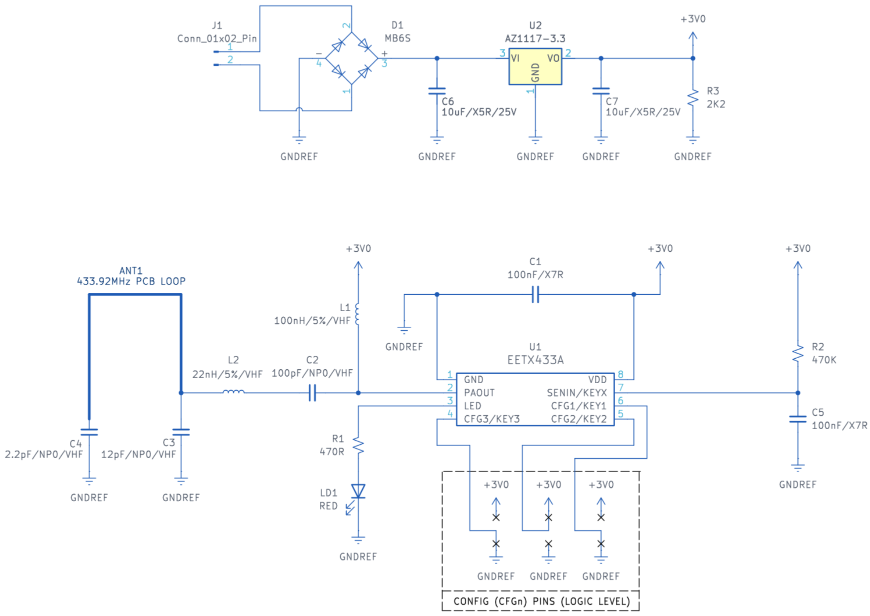


Figure 8: Typical application — transmitter installed on the high-beam headlight (bridge rectifier + 3.3 V LDO + EETX433A)

> **Vehicle-Headlight Application: to ensure a complete transmission, keep the high beam asserted for at least 2 seconds. This covers wake-up (≈ 50 ms) + PLL settling (≈ 0.2 ms) + fixed 1 200 ms transmission + margin.**

NOTES

1. Fine-tuning C3 and C4 capacitance values optimizes the antenna impedance matching network, improving transmit power and efficiency.
2. To ensure device reliability and lifetime, the recommended maximum ratings must not be exceeded. Sustained operation at these limits may cause permanent damage.
3. In applications where the input voltage may exceed the recommended VDD max, use an LDO with quiescent current ≤ 1 µA to power the device.

Data Format

The EETX433A transmits using a pulse-width OOK encoding scheme at 433.92 MHz. Each code word contains a pilot, address, data and end code. The basic time unit (λ) equals 1/3 of the bit clock; in the EETX433A, λ = 500 µs, so each bit lasts 1.5 ms.

In Transmitter mode, transmission triggers when a key connects the KEY pin to VDD (logic 1). In sensor modes (1–7), transmission triggers on the rising edge (0→1) of the SENIN pin. See "Functional Operation" for details.

Packet Structure

Each packet (≈ 50 ms) consists of: 24λ pilot + 20-bit programmable address + 4-bit data (D3 D2 D1 D0) + 4λ end code.

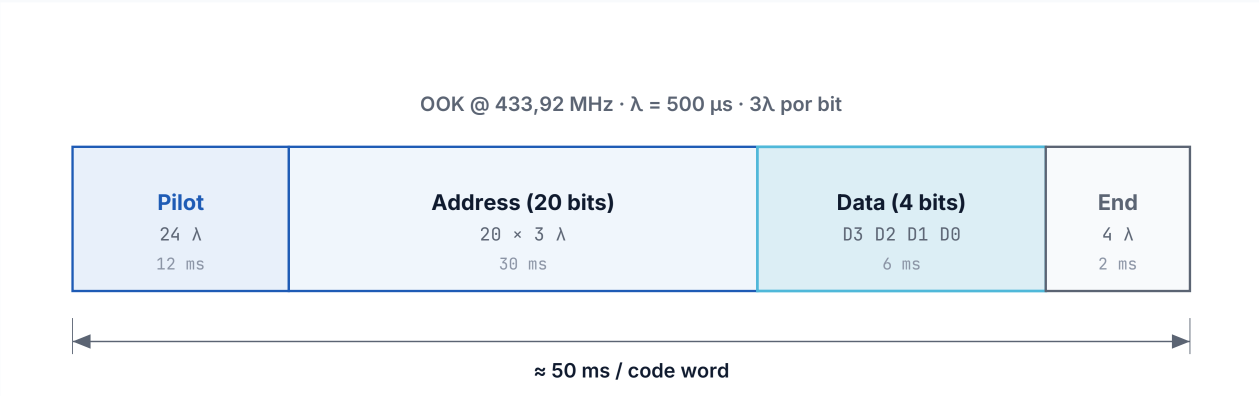


Figure 9: Transmission packet structure

Complete Frame Timing

The diagram below shows the internal fosc clock and the DOUT data line aligned across the four phases of the code word. Each address/data bit occupies 3λ; the pilot occupies 24λ and the end code 4λ.

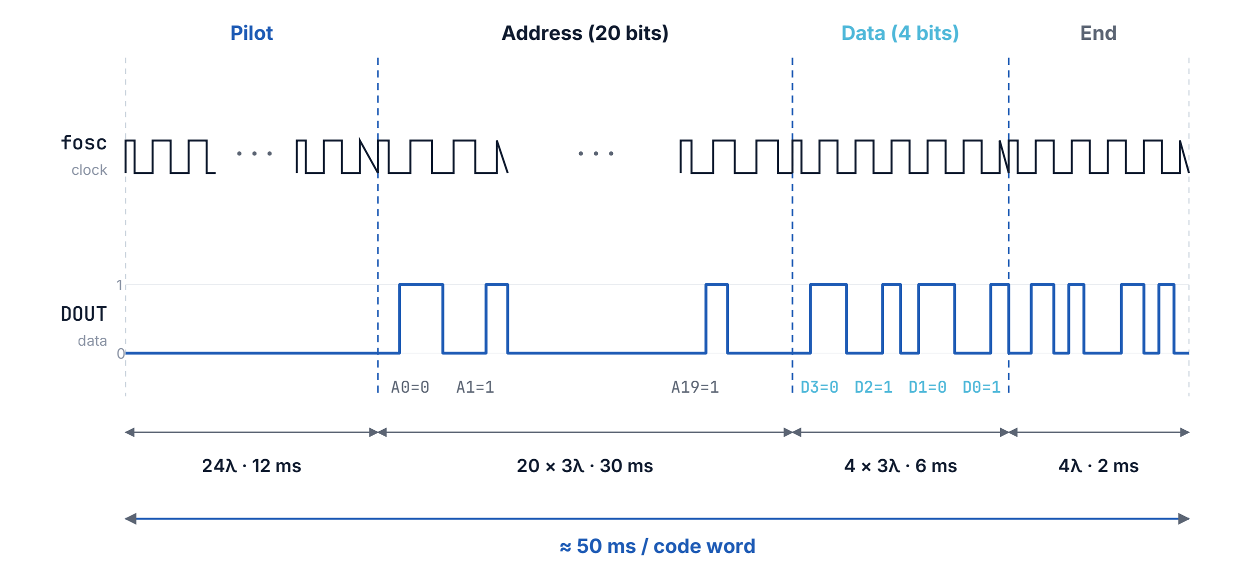


Figure 9b: Frame timing — fosc + DOUT (Pilot · Address · Data · End)

Logic Bit — Waveform

Each address/data bit occupies 3λ (1.5 ms). The logic level is set by the ratio between the low and high times:

- `Bit "0": 1λ low + 2λ high (500 μs low / 1000 μs high)`
- `Bit "1": 2λ low + 1λ high (1000 μs low / 500 μs high)`

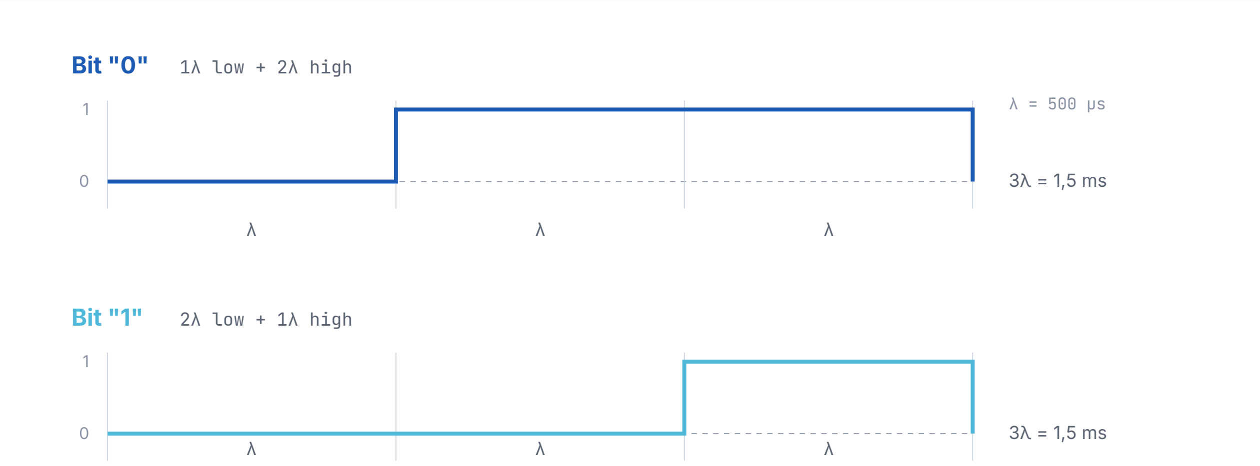


Figure 10: Logic-bit waveforms (OOK modulation)

Characteristic Times

Parameter	Símbolo	Typ.	Unit
Basic unit (λ = 1/3 of bit clock)	λ	500	µs
Bit duration (3λ)	t_BIT	1,5	ms
Pilot (24λ)	t_PIL	12	ms
Address (20 bits)	t_ADDR	30	ms
Data (4 bits)	t_DATA	6	ms
End code (4λ)	t_END	2	ms
Complete packet (typ.)	t_PKT	≈ 50	ms

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit
VDD	V_DD	2.0	3.6	V
TA	T_A	-40	85	°C
TJ	T_J	-40	150	°C
TSTG	T_STG	-50	150	°C

Recommended Operating Conditions

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
VDD	V_DD		2.1	3.0	3.3	V
Ta	T_a		-30	25	85	°C
RL	R_L			50		Ω

DC Characteristics

V_{SS}=0V, V_{DD}=3V, T_a=25°C

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
VDD	V_DD		2.0	3.0	3.6	V
IDD	I_DD	13dBm	8	12	14	mA
ISD	I_SD	KEY<0:3> floating	700	800	900	nA

Analog / RF Characteristics

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Po (433M)	P_O	RL=50Ω, VDD=3V		13		dBm
Po (433M)	P_O	RL=50Ω, VDD=3.6V		14		dBm
Δf entre unidades	Δf_unit	VDD=3V, Ta=25°C, unit-to-unit		±200	±250	kHz

Soldering Guidelines

Because this component has low thermal capacity due to its compact construction, minimize exposure to external heat. Otherwise, thermal deformation may cause damage and characteristic shifts such as frequency drift. Use a non-corrosive rosin-type flux.

1. Hand soldering (test samples)

- Use a 30 W soldering iron with tip temperature 260–300 °C.
- Maximum contact time: 5 seconds.
- Keep the tip thoroughly clean.

2. Reflow soldering (SMT)

Follow the recommended temperature profile below: preheat up to 150 °C for no more than 60 seconds, then main heating up to 210 °C for no more than 10 seconds.

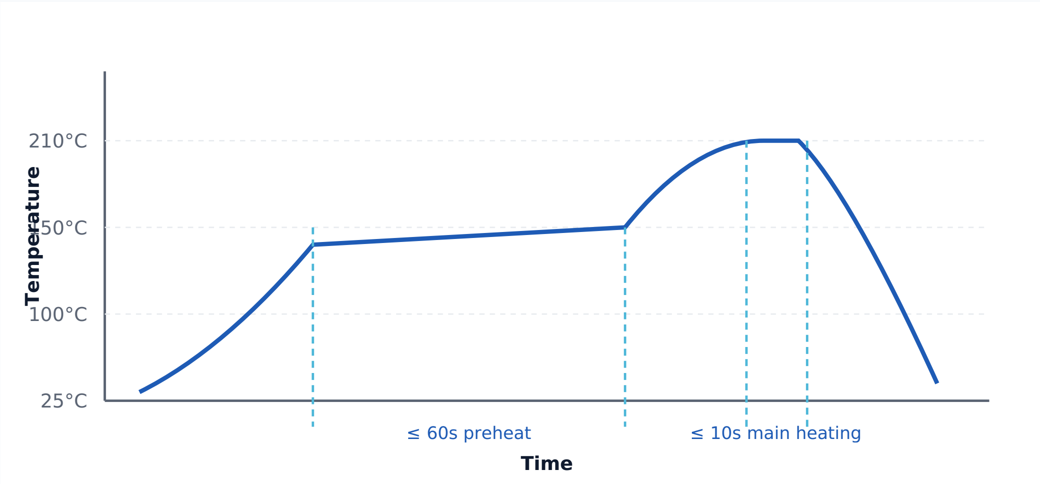


Figure 11: Recommended reflow temperature profile

Package Information

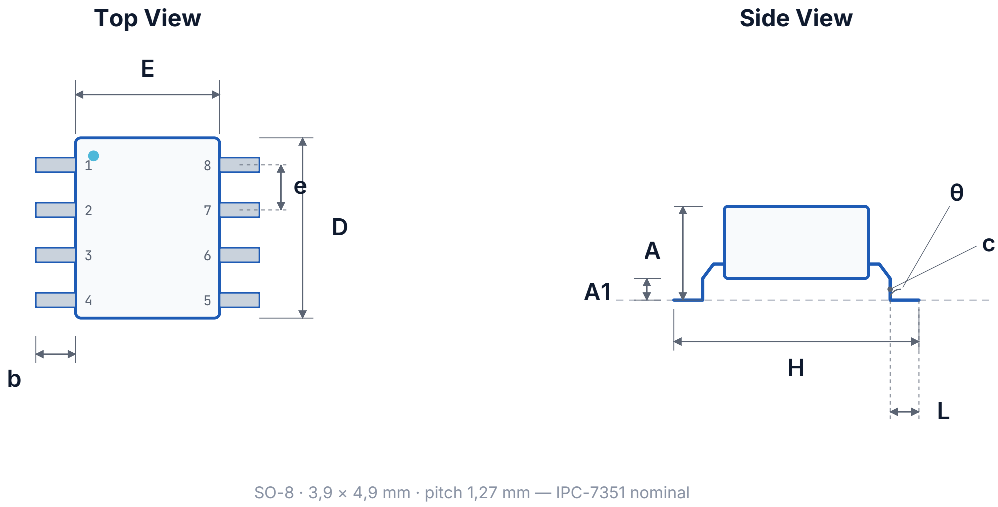


Figure 12: SOP-8 package — mechanical drawing

Symbol	Min (mm)	Nom (mm)	Max (mm)	Min (in)	Nom (in)	Max (in)
A	1.30	1.50	1.70	0.051	0.059	0.067
A1	0.06	0.16	0.26	0.002	0.006	0.010
b	0.30	0.40	0.55	0.012	0.016	0.022
c	0.15	0.25	0.35	0.006	0.010	0.014
D	4.72	4.92	5.12	0.186	0.194	0.202
E	3.75	3.95	4.15	0.148	0.156	0.163
e	–	1.27	–	–	0.050	–
H	5.70	6.00	6.30	0.224	0.236	0.248
L	0.45	0.65	0.85	0.018	0.026	0.033
θ	0°	–	8°	0°	–	8°

Packaging (Tape & Reel)

The EETX433A is supplied in carrier tape wound on a standard 330 mm reel, fully compatible with SMT pick-and-place systems.

STANDARD QUANTITY

4 000 pieces per reel

Collective packaging: 1 reel + identification card + sealed anti-static bag, per MSL-3 / J-STD-020.

Reel Dimensions — SOP-8

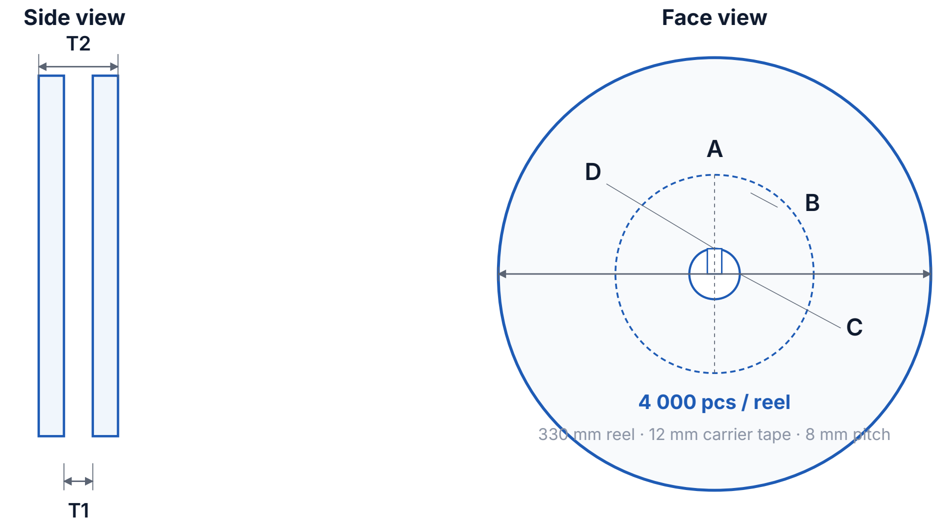


Figure 13: Standard 330 mm reel — 4 000 pcs

Symbol	Description	Dimension (mm)
A	Diâmetro externo do rolo / Reel outer diameter	330 ±1,0
B	Diâmetro interno do rolo / Reel inner diameter	62 ±1,5
C	Diâmetro do furo central / Spindle hole diameter	13,0 +0,5 / -0,2
D	Largura do entalhe-chave / Key slit width	2,0 ±0,15
T1	Espaço entre flanges / Space between flanges	12,8 +0,3 / -0,2
T2	Espessura total do rolo / Reel thickness	18,2 ±0,2

Carrier Tape Dimensions — SOP-8

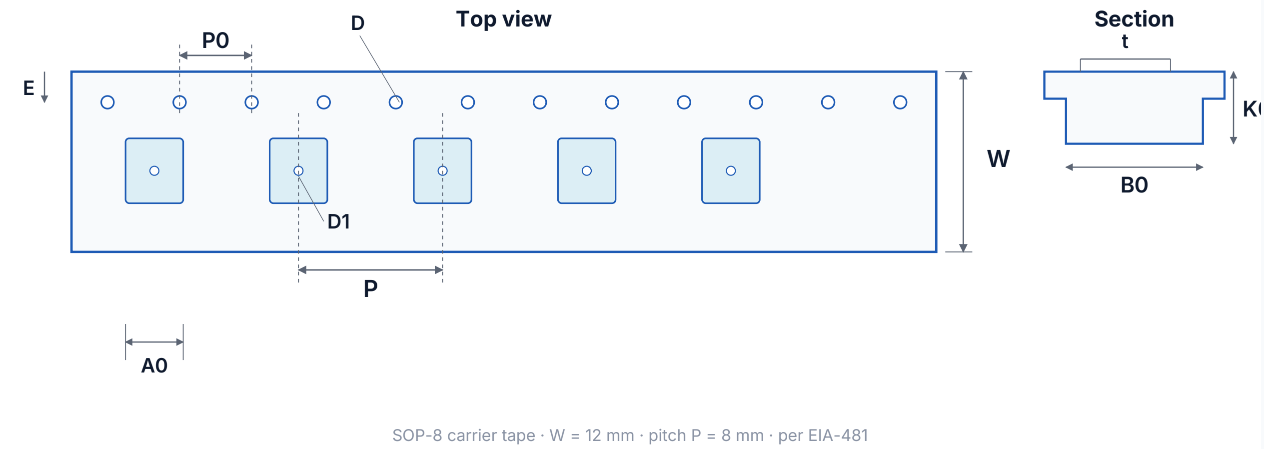


Figure 14: 12 mm carrier tape

Symbol	Description	Dimension (mm)
W	Largura da fita / Carrier tape width	12,0 +0,3 / -0,1
P	Passo dos compartimentos / Cavity pitch	8,0 ±0,1
E	Posição da perfuração / Perforation position	1,75 ±0,1
F	Cavidade → perfuração (largura) / Cavity to perf	5,5 ±0,1
D	Diâmetro da perfuração / Perforation diameter	1,55 ±0,1
D1	Diâmetro do furo da cavidade / Cavity hole diameter	1,5 +0,25
P0	Passo da perfuração / Perforation pitch	4,0 ±0,1
P1	Cavidade → perfuração (comprimento)	2,0 ±0,1
A0	Comprimento da cavidade / Cavity length	6,4 ±0,1
B0	Largura da cavidade / Cavity width	5,20 ±0,1
K0	Profundidade da cavidade / Cavity depth	2,1 ±0,1
t	Espessura da fita / Carrier tape thickness	0,3 ±0,05
C	Largura da fita de cobertura / Cover tape width	9,3

For more information about this product, visit: **https://engeletron.com.br/**

Disclaimer

Information in this document is provided in good faith and believed accurate at the time of publication. Engeletron Engenharia reserves the right to modify specifications without notice and assumes no liability for consequences arising from the use of this information.